



## Lab Report 2

### CSE3023 - Web Application and Development

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## Task 1

### ServletDataSharing

#### Login.html

```
1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4      <meta charset="UTF-8">
5      <meta name="viewport" content="width=device-width, initial-scale=1.0">
6      <title>Login Page</title>
7  </head>
8  <body>
9      <h2>Login</h2>
10     <form action="LoginServlet" method="POST">
11         <label for="username">Username:</label>
12         <input type="text" id="username" name="username"><br><br>
13
14         <label for="password">Password:</label>
15         <input type="password" id="password" name="password"><br><br>
16
17         <input type="submit" value="Login">
18     </form>
19 </body>
20 </html>
```

```
2  import java.io.IOException;
3  import java.io.PrintWriter;
4  import javax.servlet.RequestDispatcher;
5  import javax.servlet.ServletException;
6  import javax.servlet.http.HttpServlet;
7  import javax.servlet.http.HttpServletRequest;
8  import javax.servlet.http.HttpServletResponse;
9
10  /**
11   *
12   * @author Nazmi
13   */
14  public class LoginServlet extends HttpServlet {
15
16      /**
17       * Processes requests for both HTTP <code>GET</code> and <code>POST</code>
18       * methods.
19       *
20       * @param request servlet request
21       * @param response servlet response
22       * @throws ServletException if a servlet-specific error occurs
23       * @throws IOException if an I/O error occurs
24       */
25      protected void processRequest(HttpServletRequest request, HttpServletResponse response)
26          throws ServletException, IOException {
27          response.setContentType("text/html;charset=UTF-8");
28          try (PrintWriter out = response.getWriter()) {
```

```

28     try (PrintWriter out = response.getWriter()) {
29         /* TODO output your page here. You may use following sample code. */
30         out.println("<!DOCTYPE html>");
31         out.println("<html>");
32         out.println("<head>");
33         out.println("<title>Servlet LoginServlet</title>");
34         out.println("</head>");
35         out.println("<body>");
36         out.println("<h1>Servlet LoginServlet at " + request.getContextPath() + "</h1>");
37         out.println("</body>");
38         out.println("</html>");
39     }
40 }
41
42 // <editor-fold defaultstate="collapsed" desc="HttpServlet methods. Click on the + sign on the left to edit the code.">
43 /**
44  * Handles the HTTP <code>GET</code> method.
45  *
46  * @param request servlet request
47  * @param response servlet response
48  * @throws ServletException if a servlet-specific error occurs
49  * @throws IOException if an I/O error occurs
50  */
51 @Override
52 protected void doGet(HttpServletRequest request, HttpServletResponse response)
53     throws ServletException, IOException {
54     processRequest(request, response);
55 }

```

```

56
57 protected void doPost(HttpServletRequest request, HttpServletResponse
58 response) throws ServletException, IOException {
59
60     String user = request.getParameter("username");
61     String pass = request.getParameter("password");
62
63     // Semakan kata laluan (Authentication)
64     if (user.equals("Ahmad") && pass.equals("4567")) {
65
66         // Data Sharing: Menyimpan maklumat tambahan ke dalam 'request'
67         request.setAttribute("accountType", "Premium Student");
68         request.setAttribute("email", "ahmad@siswa.edu.my");
69
70         // Request Forwarding ke AccountServlet
71         RequestDispatcher rd = request.getRequestDispatcher ("AccountServlet");
72         rd.forward(request, response);
73
74     } else if (user.equals("Siti") && pass.equals("1234")) {
75         // Contoh untuk pengguna lain
76         request.setAttribute("accountType", "Standard Student");
77         request.setAttribute("email", "siti@siswa.edu.my");
78
79         RequestDispatcher rd = request.getRequestDispatcher ("AccountServlet");
80         rd.forward(request, response);
81
82     }
83 }

```

```

91     } else {
92         // Jika login gagal
93         response.setContentType("text/html");
94         PrintWriter out = response.getWriter();
95         out.println("<h3 style='color:red'>Login Failed! Invalid credentials.</h3>");
96         out.println("<a href='login.html'>Try Again</a>");
97     }
98 }
99
100 @Override
101 public String getServletInfo() {
102     return "Short description";
103 } // </editor-fold>
104
105 }
106

```

## AccountServlet

```

1
2 import java.io.IOException;
3 import java.io.PrintWriter;
4 import javax.servlet.ServletException;
5 import javax.servlet.http.HttpServlet;
6 import javax.servlet.http.HttpServletRequest;
7 import javax.servlet.http.HttpServletResponse;
8
9 /**
10  *
11  * @author Nazmi
12  */
13 public class AccountServlet extends HttpServlet {
14
15     /**
16      * Processes requests for both HTTP <code>GET</code> and <code>POST</code>
17      * methods.
18      *
19      * @param request servlet request
20      * @param response servlet response
21      * @throws ServletException if a servlet-specific error occurs
22      * @throws IOException if an I/O error occurs
23      */
24
25     protected void processRequest(HttpServletRequest request, HttpServletResponse response)
26         throws ServletException, IOException {
27         response.setContentType("text/html;charset=UTF-8");
28         try (PrintWriter out = response.getWriter()) {
29             /* TODO output your page here. You may use following sample code. */
30             out.println("<!DOCTYPE html>");
31             out.println("<html>");
32             out.println("<head>");
33             out.println("<title>Servlet AccountServlet</title>");
34             out.println("</head>");
35             out.println("<body>");
36             out.println("<h1>Servlet AccountServlet at " + request.getContextPath() + "</h1>");
37             out.println("</body>");
38             out.println("</html>");
39         }
40     }
41
42     HttpServlet methods. Click on the + sign on the left to edit the code.
43
44 }
45

```

```

66     protected void doPost(HttpServletRequest request, HttpServletResponse response)
67         throws ServletException, IOException {
68
69         response.setContentType("text/html");
70         PrintWriter out = response.getWriter();
71
72         // Ambil parameter dari form
73         String username = request.getParameter("username");
74
75         // Ambil data dari LoginServlet (request attribute)
76         String accountType = (String) request.getAttribute("accountType");
77         String email = (String) request.getAttribute("email");
78
79         out.println("<html><body>");
80         out.println("<h2>Account Information</h2>");
81         out.println("<hr>");
82         out.println("<p><strong>Welcome, " + username + "!</strong></p>");
83         out.println("<p><b>Email:</b> " + email + "</p>");
84         out.println("<p><b>Account Type:</b> " + accountType + "</p>");
85         out.println("<br>");
86         out.println("<a href='login.html'>Logout</a>");
87         out.println("</body></html>");
88     }

```

Output

## Account Information

---

**Welcome, Ahmad!**

**Email:** ahmad@siswa.edu.my

**Account Type:** Premium Student

[Logout](#)

## Task 2,3,4

```
1 USE CSE3023;
2 CREATE TABLE users (
3   id INT AUTO_INCREMENT PRIMARY KEY,
4   username VARCHAR(100),
5   password VARCHAR(100),
6   roles VARCHAR(50)
7 );
8
```

## Task 4

```
1  <!DOCTYPE html>
2  <!--
3  Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this license
4  Click nbfs://nbhost/SystemFileSystem/Templates/JSP\_Servlet/Html.html to edit this template
5  -->
6  <html>
7  <head>
8      <title>User Registration</title>
9      <meta charset="UTF-8">
10     <meta name="viewport" content="width=device-width, initial-scale=1.0">
11 </head>
12 <body>
13     <h2>Register New User</h2>
14
15     <form action="InsertServlet" method="post">
16
17         <label>Username:</label><br>
18         <input type="text" name="username" required><br><br>
19
20         <label>Password:</label><br>
21         <input type="password" name="password" required><br><br>
22
23         <label>Role:</label><br>
24         <select name="roles">
25
26             <option value="admin">Admin</option>
27             <option value="user">User</option>
28         </select><br><br>
29
30         <input type="submit" value="Submit">
31
32     </form>
33 </body>
34 </html>
```

## InsertServlet

```
3 import java.io.IOException;
4 import java.io.PrintWriter;
5 import javax.servlet.ServletException;
6 import javax.servlet.http.HttpServlet;
7 import javax.servlet.http.HttpServletRequest;
8 import javax.servlet.http.HttpServletResponse;
9 import java.sql.Connection;
10 import java.sql.DriverManager;
11 import java.sql.PreparedStatement;
12 import javax.servlet.annotation.WebServlet;
13
14 @WebServlet("/InsertServlet")
15 public class InsertServlet extends HttpServlet {
16
17     protected void doPost(HttpServletRequest request, HttpServletResponse response) throws ServletException, IOException {
18         String username = request.getParameter("username");
19         String password = request.getParameter("password");
20         String roles = request.getParameter("roles");
21         try {
22             Class.forName("com.mysql.cj.jdbc.Driver");
23             Connection conn = DriverManager.getConnection("jdbc:mysql://localhost:3306/CSE3023", "root", "admin");
24             String sql = "INSERT INTO users (username, password, roles) VALUES(?, ?, ?)";
25             PreparedStatement pstmt = conn.prepareStatement(sql);
26             pstmt.setString(1, username);
27             pstmt.setString(2, password);
28             pstmt.setString(3, roles);
29             pstmt.executeUpdate();
30             pstmt.close();
31             conn.close();
32             response.sendRedirect("ViewServlet");
33         } catch (Exception e) {
34             e.printStackTrace();
35         }
36     }
37 }
```

## viewServlet

```
3 import java.io.IOException;
4 import java.io.PrintWriter;
5 import java.sql.Connection;
6 import java.sql.DriverManager;
7 import java.sql.ResultSet;
8 import java.sql.Statement;
9 import javax.servlet.ServletException;
10 import javax.servlet.annotation.WebServlet;
11 import javax.servlet.http.HttpServlet;
12 import javax.servlet.http.HttpServletRequest;
13 import javax.servlet.http.HttpServletResponse;
14
15 @WebServlet("/ViewServlet")
16 public class ViewServlet extends HttpServlet {
17
18     @Override
19     protected void doGet(HttpServletRequest request, HttpServletResponse response) throws ServletException, IOException {
20         response.setContentType("text/html");
21         PrintWriter out = response.getWriter();
22
23         out.println("<h2>User List</h2>");
24         out.println("<table border='1'><tr><th>ID</th><th>Username</th><th>Password</th> <th>Role</th> <th>Actions</th> </tr>");
25
26         try {
27             Class.forName("com.mysql.cj.jdbc.Driver");
28             Connection conn = DriverManager.getConnection(
29                 "jdbc:mysql://localhost:3306 / CSE3023", "root", "admin");
30             Statement stmt = conn.createStatement();
31             ResultSet rs = stmt.executeQuery("SELECT * FROM users");
32             while (rs.next()) {
33                 int id = rs.getInt("id");
34                 out.println("<td>");
35                 out.println("<td>" + id + "</td>");
36                 out.println("<td>" + rs.getString("username") + "</td>");
37             }
38             // Untuk lab, kita paparkan password. Untuk sistem sebenar, Jangan paparkan!
39         }
40     }
41 }
```

```

38 out.println("<td>" + rs.getString("password") + "</td>");
39 out.println("<td>" + rs.getString("roles") + "</td>");
40 out.println("<td><a href='UpdateServlet?id=" + id + "'>Edit</a> |");
41 out.println("<a href='DeleteServlet?id=" + id + "'>Delete</a> </td>");
42 out.println("</tr>");
43 }
44 out.println("</table>");
45 out.println("<br><a href='index.html'>Add New User</a>");
46 rs.close();
47 stmt.close();
48 conn.close();
49 } catch (Exception e) {
50 e.printStackTrace();
51 }
52 }
53 }
54 }
55

```

## deleteServlet

```

1 package com.lab.controller;
2
3 import java.sql.Connection;
4 import java.sql.DriverManager;
5 import java.sql.PreparedStatement;
6 import java.io.IOException;
7 import javax.servlet.ServletException;
8 import javax.servlet.annotation.WebServlet;
9 import javax.servlet.http.HttpServlet;
10 import javax.servlet.http.HttpServletRequest;
11 import javax.servlet.http.HttpServletResponse;
12
13 @WebServlet("/DeleteServlet")
14 public class DeleteServlet extends HttpServlet {
15
16     protected void doGet(HttpServletRequest request, HttpServletResponse response) throws ServletException, IOException {
17         String id = request.getParameter("id");
18         try {
19             Class.forName("com.mysql.cj.jdbc.Driver");
20             Connection conn = (Connection) DriverManager.getConnection("jdbc:mysql://localhost:3306 /CSE3023", "root", "admin");
21             String sql = "DELETE FROM users WHERE id=?";
22             PreparedStatement pstmt = conn.prepareStatement(sql);
23             pstmt.setInt(1, Integer.parseInt(id));
24             pstmt.executeUpdate();
25             pstmt.close();
26             conn.close();
27             response.sendRedirect("ViewServlet");
28         } catch (Exception e) {
29             e.printStackTrace();
30         }
31     }
32 }

```



## UpdateServlet

```
3 import java.io.IOException;
4 import java.io.PrintWriter;
5 import java.sql.Connection;
6 import java.sql.DriverManager;
7 import java.sql.PreparedStatement;
8 import java.sql.ResultSet;
9 import javax.servlet.ServletException;
10 import javax.servlet.annotation.WebServlet;
11 import javax.servlet.http.HttpServlet;
12 import javax.servlet.http.HttpServletRequest;
13 import javax.servlet.http.HttpServletResponse;
14
15 @WebServlet("/UpdateServlet")
16 public class UpdateServlet extends HttpServlet {
17
18     // Display existing record
19     protected void doGet(HttpServletRequest request,
20         HttpServletResponse response)
21         throws ServletException, IOException {
22
23         response.setContentType("text/html");
24         PrintWriter out = response.getWriter();
25
26         String id = request.getParameter("id");
27
28         try {
29
30             Class.forName("com.mysql.cj.jdbc.Driver");
31
32             Connection conn = DriverManager.getConnection(
33                 "jdbc:mysql://localhost:3306/CSE3023",
34                 "root",
35                 "admin");
36
37             String sql = "SELECT * FROM users WHERE id=?";
38
39             PreparedStatement pstmt = conn.prepareStatement(sql);
40             pstmt.setInt(1, Integer.parseInt(id));
41
42             ResultSet rs = pstmt.executeQuery();
43
44             if (rs.next()) {
45
46                 String currentRole = rs.getString("roles");
47
48                 out.println("<h2>Update User</h2>");
49
50                 out.println("<form action='UpdateServlet' method='POST'>");
51
52                 out.println("<input type='hidden' name='id' value='"
53                     + rs.getInt("id") + "'>");
54
55                 out.println("Username: ");
56                 out.println("<input type='text' name='username' value='"
57                     + rs.getString("username")
58                     + "' required><br><br>");
59
60                 out.println("Password: ");
61                 out.println("<input type='text' name='password' value='"
62                     + rs.getString("password")
63                     + "' required><br><br>");
64
65                 out.println("Role: ");
66                 out.println("<select name='roles'>");
67
68                 out.println("<option value='Admin' "
69                     + (currentRole.equals("Admin")
70                     ? "selected" : "")
71                     + ">Admin</option>");
```

```
73 out.println("<option value='Staff' "
74 + (currentRole.equals("Staff")
75 ? "selected" : "")
76 + ">Staff</option>");
77
78 out.println("<option value='Student' "
79 + (currentRole.equals("Student")
80 ? "selected" : "")
81 + ">Student</option>");
82
83 out.println("</select><br><br>");
84
85 out.println("<input type='submit' value='Update User'>");
86
87 out.println("</form>");
88 }
89
90 rs.close();
91 pstmt.close();
92 conn.close();
93
94 } catch (Exception e) {
95     e.printStackTrace();
96 }
97
98 // Save updated record
99 protected void doPost(HttpServletRequest request,
100 HttpServletResponse response)
101     throws ServletException, IOException {
102
103     String id = request.getParameter("id");
104     String username = request.getParameter("username");
105     String password = request.getParameter("password");
106     String roles = request.getParameter("roles");
107
108
109     try {
110
111         Class.forName("com.mysql.cj.jdbc.Driver");
112
113         Connection conn = DriverManager.getConnection(
114             "jdbc:mysql://localhost:3306/CSE3023",
115             "root",
116             "admin");
117
118         String sql =
119             "UPDATE users SET username=?, password=?, roles=? WHERE
120
121         PreparedStatement pstmt = conn.prepareStatement(sql);
122
123         pstmt.setString(1, username);
124         pstmt.setString(2, password);
125         pstmt.setString(3, roles);
126         pstmt.setInt(4, Integer.parseInt(id));
127
128         pstmt.executeUpdate();
129
130         pstmt.close();
131         conn.close();
132
133         response.sendRedirect("ViewServlet");
134
135     } catch (Exception e) {
136         e.printStackTrace();
137     }
138 }
139 }
```

Output:

## Register New User

Username:

Password:

Role:

# User List

< / tr >

| ID | Username | Password | Role  | Actions                                                |
|----|----------|----------|-------|--------------------------------------------------------|
| 1  | Nazmi    | Nazmi123 | Admin | <a href="#">Edit</a>   <a href="#">Delete</a> < / td > |

[Add New User](#)

< / tr >

| ID | Username | Password | Role  | Actions                                                |
|----|----------|----------|-------|--------------------------------------------------------|
| 1  | Nazmi    | Nazmi123 | Admin | <a href="#">Edit</a>   <a href="#">Delete</a> < / td > |
| 11 | Qayyum   | Nazmi123 | admin | <a href="#">Edit</a>   <a href="#">Delete</a> < / td > |
| 12 | Malik    | Nazmi123 | user  | <a href="#">Edit</a>   <a href="#">Delete</a> < / td > |
| 13 | baiti    | Nazmi123 | admin | <a href="#">Edit</a>   <a href="#">Delete</a> < / td > |
| 14 | izz      | Nazmi123 | admin | <a href="#">Edit</a>   <a href="#">Delete</a> < / td > |

[Add New User](#)

## Task 5

### InsertServlet

```
1 package com.lab.controller;
2
3 import com.lab.dao.UserDAO;
4 import com.lab.model.User;
5 import java.io.IOException;
6 import javax.servlet.ServletException;
7 import javax.servlet.http.*;
8
9 public class InsertServlet extends HttpServlet {
10
11     private UserDAO userDAO;
12
13     public void init() {
14         userDAO = new UserDAO();
15     }
16
17     protected void doPost(HttpServletRequest request,
18         HttpServletResponse response)
19         throws ServletException, IOException {
20
21         String username = request.getParameter("username");
22         String password = request.getParameter("password");
23         String roles = request.getParameter("roles");
24
25         User user = new User(username, password, roles);
26
27         userDAO.insertUser(user);
28
29         response.sendRedirect("ViewServlet");
30     }
31 }
```

## Viewservlet

```
1 package com.lab.controller;
2
3 import com.lab.dao.UserDAO;
4 import com.lab.model.User;
5 import java.io.IOException;
6 import java.io.PrintWriter;
7 import java.util.List;
8 import javax.servlet.ServletException;
9 import javax.servlet.http.*;
10
11 public class ViewServlet extends HttpServlet {
12
13     private UserDAO userDAO;
14
15     public void init() {
16         userDAO = new UserDAO();
17     }
18
19     protected void doGet(HttpServletRequest request,
20                           HttpServletResponse response)
21         throws ServletException, IOException {
22
23         response.setContentType("text/html");
24
25         PrintWriter out = response.getWriter();
26
27         List<User> listUser = userDAO.selectAllUsers();
28
29         out.println("<h2>User List (Using DAO)</h2>");
30
31         out.println("<table border='1'>");
32         out.println("<tr>");
33         out.println("<th>ID</th>");
34         out.println("<th>Username</th>");
35         out.println("<th>Password</th>");
36         out.println("<th>Role</th>");
37         out.println("<th>Action</th>");
38         out.println("</tr>");
39
40         for (User user : listUser) {
41
42             out.println("<tr>");
43             out.println("<td>" + user.getId() + "</td>");
44             out.println("<td>" + user.getUsername() + "</td>");
45             out.println("<td>" + user.getPassword() + "</td>");
46             out.println("<td>" + user.getRoles() + "</td>");
47
48             out.println("<td>");
49             out.println("<a href='UpdateServlet?id="
50                 + user.getId()
51                 + "'>Edit</a> | ");
52
53             out.println("<a href='DeleteServlet?id="
54                 + user.getId()
55                 + "'>Delete</a>");
56             out.println("</td>");
57
58             out.println("</tr>");
59         }
60
61         out.println("</table>");
62
63         out.println("<br>");
64         out.println("<a href='index.html'>Add New User</a>");
65     }
66 }
```

## DeleteServlet

```
1 package com.lab.controller;
2
3 import com.lab.dao.UserDAO;
4 import java.io.IOException;
5 import javax.servlet.ServletException;
6 import javax.servlet.http.*;
7
8 public class DeleteServlet extends HttpServlet {
9
10     private UserDAO userDAO;
11
12     public void init() {
13         userDAO = new UserDAO();
14     }
15
16     protected void doGet(HttpServletRequest request,
17                           HttpServletResponse response)
18         throws ServletException, IOException {
19
20         int id =
21             Integer.parseInt(
22                 request.getParameter("id"));
23
24         userDAO.deleteUser(id);
25
26         response.sendRedirect("ViewServlet");
27     }
28 }
```

## UpdateServlet

```
3 import com.lab.dao.UserDAO;
4 import com.lab.model.User;
5 import java.io.IOException;
6 import java.io.PrintWriter;
7 import javax.servlet.ServletException;
8 import javax.servlet.http.*;
9
10 public class UpdateServlet extends HttpServlet {
11
12     private UserDAO userDAO;
13
14     public void init() {
15         userDAO = new UserDAO();
16     }
17
18     protected void doGet(HttpServletRequest request,
19                           HttpServletResponse response)
20         throws ServletException, IOException {
21
22         response.setContentType("text/html");
23
24         PrintWriter out = response.getWriter();
25
26         int id =
27             Integer.parseInt(
28                 request.getParameter("id"));
29
30         User user = userDAO.selectUser(id);
31
32         if (user != null) {
33             String currentRole = user.getRoles();
34
35             out.println("<h2>Update User</h2>");
36
37             out.println("<form action='UpdateServlet' method='POST'>");
38
39             out.println("<input type='hidden' name='id' value='"
40                         + user.getId() + "'>");
41
42             out.println("Username: <input type='text' name='username' value='"
43                         + user.getUsername() + "' required><br><br>");
44
45             out.println("Password: <input type='text' name='password' value='"
46                         + user.getPassword() + "' required><br><br>");
47
48             out.println("Role: <select name='roles'>");
49
50             out.println("<option value='Admin' "
51                         + (currentRole.equals("Admin") ? "selected" : "")
52                         + ">Admin</option>");
53
54             out.println("<option value='Staff' "
55                         + (currentRole.equals("Staff") ? "selected" : "")
56                         + ">Staff</option>");
57
58             out.println("<option value='Student' "
59                         + (currentRole.equals("Student") ? "selected" : "")
60                         + ">Student</option>");
61
62             out.println("</select><br><br>");
63
64             out.println("<input type='submit' value='Update User'>");
65
66             out.println("</form>");
67         }
68     }
69 }
```

```

72     protected void doPost(HttpServletRequest request,
73                            HttpServletResponse response)
74         throws ServletException, IOException {
75
76         int id =
77             Integer.parseInt(
78                 request.getParameter("id"));
79
80         String username =
81             request.getParameter("username");
82
83         String password =
84             request.getParameter("password");
85
86         String roles =
87             request.getParameter("roles");
88
89         User user =
90             new User(id, username, password, roles);
91
92         userDao.updateUser(user);
93
94         response.sendRedirect("ViewServlet");
95     }

```

## UserDAO

```

3  import com.lab.model.User;
4  import java.sql.Connection;
5  import java.sql.DriverManager;
6  import java.sql.PreparedStatement;
7  import java.sql.ResultSet;
8  import java.sql.SQLException;
9  import java.util.ArrayList;
10 import java.util.List;
11
12 public class UserDao {
13
14     private String jdbcURL = "jdbc:mysql://localhost:3306/CSE3023";
15     private String jdbcUsername = "root";
16     private String jdbcPassword = "admin";
17
18     protected Connection getConnection() {
19
20         Connection connection = null;
21
22         try {
23
24             Class.forName("com.mysql.cj.jdbc.Driver");
25
26             connection = DriverManager.getConnection(
27                 jdbcURL,
28                 jdbcUsername,
29                 jdbcPassword);
30
31         } catch (Exception e) {
32             e.printStackTrace();
33         }
34
35         return connection;
36     }

```

```

38 // INSERT
39 public void insertUser(User user) {
40
41     String sql =
42         "INSERT INTO users(username,password,roles) VALUES(?,?,?)";
43
44     try (Connection conn = getConnection();
45         PreparedStatement pstmt = conn.prepareStatement(sql)) {
46
47         pstmt.setString(1, user.getUsername());
48         pstmt.setString(2, user.getPassword());
49         pstmt.setString(3, user.getRoles());
50
51         pstmt.executeUpdate();
52
53     } catch (SQLException e) {
54         e.printStackTrace();
55     }
56 }
57
58 // SELECT ONE
59 public User selectUser(int id) {
60
61     User user = null;
62
63     String sql =
64         "SELECT * FROM users WHERE id=?";
65
66     try (Connection conn = getConnection();
67         PreparedStatement pstmt = conn.prepareStatement(sql)) {
68
69         pstmt.setInt(1, id);
70
71         ResultSet rs = pstmt.executeQuery();

```

```

73         if (rs.next()) {
74
75             String username = rs.getString("username");
76             String password = rs.getString("password");
77             String roles = rs.getString("roles");
78
79             user = new User(id, username, password, roles);
80         }
81
82     } catch (SQLException e) {
83         e.printStackTrace();
84     }
85
86     return user;
87 }
88
89 // SELECT ALL
90 public List<User> selectAllUsers() {
91
92     List<User> users = new ArrayList<>();
93
94     String sql = "SELECT * FROM users";
95
96     try (Connection conn = getConnection();
97         PreparedStatement pstmt = conn.prepareStatement(sql)) {
98
99         ResultSet rs = pstmt.executeQuery();
100
101         while (rs.next()) {
102
103             int id = rs.getInt("id");
104             String username = rs.getString("username");
105             String password = rs.getString("password");
106             String roles = rs.getString("roles");

```

```

108         users.add(
109             new User(
110                 id,
111                 username,
112                 password,
113                 roles
114             )
115         );
116     }
117 }
118 } catch (SQLException e) {
119     e.printStackTrace();
120 }
121 }
122 return users;
123 }
124
125 // UPDATE
126 public boolean updateUser(User user) {
127
128     boolean rowUpdated = false;
129
130     String sql =
131         "UPDATE users SET username=?, password=?, roles=? WHERE id=?";
132
133     try (Connection conn = getConnection();
134         PreparedStatement pstmt = conn.prepareStatement(sql)) {
135
136         pstmt.setString(1, user.getUsername());
137         pstmt.setString(2, user.getPassword());
138         pstmt.setString(3, user.getRoles());
139         pstmt.setInt(4, user.getId());
140
141         rowUpdated = pstmt.executeUpdate() > 0;

```

```

143     } catch (SQLException e) {
144         e.printStackTrace();
145     }
146 }
147 return rowUpdated;
148 }
149
150 // DELETE
151 public boolean deleteUser(int id) {
152
153     boolean rowDeleted = false;
154
155     String sql =
156         "DELETE FROM users WHERE id=?";
157
158     try (Connection conn = getConnection();
159         PreparedStatement pstmt = conn.prepareStatement(sql)) {
160
161         pstmt.setInt(1, id);
162
163         rowDeleted = pstmt.executeUpdate() > 0;
164
165     } catch (SQLException e) {
166         e.printStackTrace();
167     }
168
169     return rowDeleted;
170 }
171 }

```



## User.java

```
1  package com.lab.model;
2
3  public class User {
4
5      private int id;
6      private String username;
7      private String password;
8      private String roles;
9
10     // Default Constructor
11     public User() {
12     }
13
14     // Constructor with ID
15     public User(int id, String username, String password, String roles)
16     {
17         this.id = id;
18         this.username = username;
19         this.password = password;
20         this.roles = roles;
21     }
22
23     // Constructor without ID
24     public User(String username, String password, String roles) {
25         this.username = username;
26         this.password = password;
27         this.roles = roles;
28     }
29
30     public int getId() {
31         return id;
32     }
33
34     public void setId(int id) {
35         this.id = id;
36     }
```

```
36
37     public String getUsername() {
38         return username;
39     }
40
41     public void setUsername(String username) {
42         this.username = username;
43     }
44
45     public String getPassword() {
46         return password;
47     }
48
49     public void setPassword(String password) {
50         this.password = password;
51     }
52
53     public String getRoles() {
54         return roles;
55     }
56
57     public void setRoles(String roles) {
58         this.roles = roles;
59     }
60 }
```

# User List (Using DAO)

| ID | Username | Password | Role  | Action                                        |
|----|----------|----------|-------|-----------------------------------------------|
| 1  | Nazmi    | Nazmi123 | Admin | <a href="#">Edit</a>   <a href="#">Delete</a> |
| 11 | Qayyum   | Nazmi123 | admin | <a href="#">Edit</a>   <a href="#">Delete</a> |
| 12 | Malik    | Nazmi123 | user  | <a href="#">Edit</a>   <a href="#">Delete</a> |
| 13 | baiti    | Nazmi123 | admin | <a href="#">Edit</a>   <a href="#">Delete</a> |
| 14 | izz      | Nazmi123 | admin | <a href="#">Edit</a>   <a href="#">Delete</a> |
| 16 | Kucing   | Nazmi123 | Admin | <a href="#">Edit</a>   <a href="#">Delete</a> |

## [Add New User](#)

Exercise

Add\_html.html

```
1 <!DOCTYPE html>
2 <!--
3 Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this license
4 Click nbfs://nbhost/SystemFileSystem/Templates/JSP\_Servlet/Html.html to edit this template
5 -->
6 <html>
7 <head>
8 <title>Add Product</title>
9 <meta charset="UTF-8">
10 </head>
11 <body>
12 <h2>Add New Product</h2>
13 <form action="InsertProductServlet" method="POST">
14 <label>Name:</label><br>
15 <input type="text" name="name" required><br><br>
16
17 <label>Category:</label><br>
18 <input type="text" name="category" required><br><br>
19
20 <label>Price:</label><br>
21 <input type="number" name="price" step="0.01" required><br><br>
22
23 <label>Quantity:</label><br>
24 <input type="number" name="quantity" required><br><br>
25
26 <input type="submit" value="Add Product">
27 <form action="insert" method="post">
28 </form>
29 </body>
30 </html>
```

## Index.html

```
1 <!DOCTYPE html>
2 <!--
3 Click nbfs://nbhost/SystemFileSystem/Templates/licenses/license-default.txt to change this license
4 Click nbfs://nbhost/SystemFileSystem/Templates/JSP\_Servlet/Html.html to edit this template
5 -->
6 <html>
7 <head>
8 <title>Add Product</title>
9 <meta charset="UTF-8">
10 </head>
11 <body>
12 <h2>Add New Product</h2>
13 <form action="InsertProductServlet" method="POST">
14 <label>Name:</label><br>
15 <input type="text" name="name" required><br><br>
16
17 <label>Category:</label><br>
18 <input type="text" name="category" required><br><br>
19
20 <label>Price:</label><br>
21 <input type="number" name="price" step="0.01" required><br><br>
22
23 <label>Quantity:</label><br>
24 <input type="number" name="quantity" required><br><br>
25
26 <input type="submit" value="Add Product">
27 </form>
28 </body>
29 </html>
```

## InsertProductServlet

```
63 protected void doGet(HttpServletRequest request, HttpServletResponse response)
64     throws ServletException, IOException {
65     processRequest(request, response);
66 }
67 /**
68  * Handles the HTTP POST method.
69  *
70  * @param request servlet request
71  * @param response servlet response
72  * @throws ServletException if a servlet-specific error occurs
73  * @throws IOException if an I/O error occurs
74  */
75 @Override
76 protected void doPost(HttpServletRequest request, HttpServletResponse response)
77     throws ServletException, IOException {
78
79     String name = request.getParameter("name");
80     String category = request.getParameter("category");
81     double price = Double.parseDouble(request.getParameter("price"));
82     int quantity = Integer.parseInt(request.getParameter("quantity"));
83
84     Product product = new Product(name, category, price, quantity);
85     productDAO.insertProduct(product);
86
87
88     response.sendRedirect("ViewProductServlet");
89 }
90 }
```

## UpdateProductServlet

```
67     @Override
68     protected void doGet(HttpServletRequest request, HttpServletResponse response)
69         throws ServletException, IOException {
70         processRequest(request, response);
71     }
72
73     /**
74      * Handles the HTTP <code>POST</code> method.
75      *
76      * @param request servlet request
77      * @param response servlet response
78      * @throws ServletException if a servlet-specific error occurs
79      * @throws IOException if an I/O error occurs
80      */
81     @Override
82     protected void doPost(HttpServletRequest request, HttpServletResponse response)
83         throws ServletException, IOException {
84         int id = Integer.parseInt(request.getParameter("id"));
85         String name = request.getParameter("name");
86         String category = request.getParameter("category");
87         double price = Double.parseDouble(request.getParameter("price"));
88         int quantity = Integer.parseInt(request.getParameter("quantity"));
89
90         Product product = new Product(id, name, category, price, quantity);
91
92         productDAO.updateProduct(product);
93
94         response.sendRedirect("view");
95     }
```

## DeleteProductservlet

```
17
18     * @author Nazmi
19     */
20     public class DeleteProductServlet extends HttpServlet {
21         private ProductDAO productDAO;
22
23         public void init() {
24             productDAO = new ProductDAO();
25         }
26
27         /**
28          * Processes requests for both HTTP <code>GET</code> and <code>POST</code>
29          * methods.
30          *
31          * @param request servlet request
32          * @param response servlet response
33          * @throws ServletException if a servlet-specific error occurs
34          * @throws IOException if an I/O error occurs
35          */
36         protected void processRequest(HttpServletRequest request, HttpServletResponse response)
37             throws ServletException, IOException {
38             response.setContentType("text/html;charset=UTF-8");
39             try (PrintWriter out = response.getWriter()) {
40                 /* TODO output your page here. You may use following sample code. */
41                 out.println("<!DOCTYPE html>");
42                 out.println("<html>");
43                 out.println("<head>");
44                 out.println("<title>Servlet DeleteProductServlet</title>");
45                 out.println("</head>");
46                 out.println("<body>");
47                 out.println("<h1>Servlet DeleteProductServlet at " + request.getContextPath() + "</h1>");
48                 out.println("</body>");
49                 out.println("</html>");
50             }
51     }
```

```

62     @Override
63     protected void doGet(HttpServletRequest request, HttpServletResponse response)
64         throws ServletException, IOException {
65
66         int id = Integer.parseInt(request.getParameter("id"));
67
68         productDAO.deleteProduct(id);
69
70         response.sendRedirect("view");
71     }
72
73
74     /**
75      * Handles the HTTP <code>POST</code> method.
76      *
77      * @param request servlet request
78      * @param response servlet response
79      * @throws ServletException if a servlet-specific error occurs
80      * @throws IOException if an I/O error occurs
81      */
82     @Override
83     protected void doPost(HttpServletRequest request, HttpServletResponse response)
84         throws ServletException, IOException {
85         processRequest(request, response);
86     }
87
88     /**
89      * Returns a short description of the servlet.
90      *
91      * @return a String containing servlet description
92      */
93     @Override
94     public String getServletInfo() {
95         return "Short description";
96     } // </editor-fold>

```

## ProductDAO

```

/*
 * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this
license
 * Click nbfs://nbhost/SystemFileSystem/Templates/Classes/Class.java to edit this template
 */
/*
 * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this
license
 * Click nbfs://nbhost/SystemFileSystem/Templates/Classes/Class.java to edit this template
 */

```

```
package com.lab.dao;
```

```
import java.sql.*;
import java.util.*;
import com.lab.model.Product;
```

```
public class ProductDAO {
```

```

    private String jdbcURL = "jdbc:mysql://localhost:3306/CSE3023";
    private String jdbcUsername = "root";
    private String jdbcPassword = "admin";

```

```
    private static final String INSERT_SQL =
```

```
"INSERT INTO products (name, category, price, quantity) VALUES (?, ?, ?, ?)";
```

```
private static final String SELECT_ALL =  
    "SELECT * FROM products";
```

```
private static final String DELETE_SQL =  
    "DELETE FROM products WHERE id=?";
```

```
private static final String UPDATE_SQL =  
    "UPDATE products SET name=?, category=?, price=?, quantity=? WHERE id=?";
```

```
protected Connection getConnection() {  
    Connection con = null;  
    try {  
        Class.forName("com.mysql.cj.jdbc.Driver");  
        con = DriverManager.getConnection(jdbcURL, jdbcUsername, jdbcPassword);  
    } catch (Exception e) {  
        e.printStackTrace();  
    }  
    return con;  
}
```

```
// INSERT
```

```
public void insertProduct(Product p) {  
    try (Connection con = getConnection();  
        PreparedStatement ps = con.prepareStatement(INSERT_SQL)) {  
  
        ps.setString(1, p.getName());  
        ps.setString(2, p.getCategory());  
        ps.setDouble(3, p.getPrice());  
        ps.setInt(4, p.getQuantity());  
  
        ps.executeUpdate();  
    } catch (Exception e) {  
        e.printStackTrace();  
    }  
}
```

```
// SELECT
```

```
public List<Product> selectAllProducts() {  
    List<Product> list = new ArrayList<>();  
  
    try (Connection con = getConnection();  
        PreparedStatement ps = con.prepareStatement(SELECT_ALL);
```

```

        ResultSet rs = ps.executeQuery() {

        while (rs.next()) {
            Product p = new Product(
                rs.getInt("id"),
                rs.getString("name"),
                rs.getString("category"),
                rs.getDouble("price"),
                rs.getInt("quantity")
            );
            list.add(p);
        }
    } catch (Exception e) {
        e.printStackTrace();
    }

    return list;
}

// DELETE
public void deleteProduct(int id) {
    try (Connection con = getConnection();
        PreparedStatement ps = con.prepareStatement(DELETE_SQL)) {

        ps.setInt(1, id);
        ps.executeUpdate();
    } catch (Exception e) {
        e.printStackTrace();
    }
}

// UPDATE
public void updateProduct(Product p) {
    try (Connection con = getConnection();
        PreparedStatement ps = con.prepareStatement(UPDATE_SQL)) {

        ps.setString(1, p.getName());
        ps.setString(2, p.getCategory());
        ps.setDouble(3, p.getPrice());
        ps.setInt(4, p.getQuantity());
        ps.setInt(5, p.getId());

        ps.executeUpdate();
    } catch (Exception e) {

```

```
        e.printStackTrace();
    }
}

}

Product
```



```
1 package com.lab.model;
2
3 /**
4  *
5  * @author Nazmi
6  */
7 public class Product {
8
9     private int id;
10    private String name;
11    private String category;
12    private double price;
13    private int quantity;
14
15    public Product() {}
16
17    public Product(int id, String name, String category, double price, int quantity) {
18        this.id = id;
19        this.name = name;
20        this.category = category;
21        this.price = price;
22        this.quantity = quantity;
23    }
24
25    public Product(String name, String category, double price, int quantity) {
26        this.name = name;
27        this.category = category;
28        this.price = price;
29        this.quantity = quantity;
30    }
31
32    public int getId() {
33        return id;
34    }
```

```
36 [-] public void setId(int id) {
37     |     this.id = id;
38     | }
39
40 [-] public String getName() {
41     |     return name;
42     | }
43
44 [-] public void setName(String name) {
45     |     this.name = name;
46     | }
47
48 [-] public String getCategory() {
49     |     return category;
50     | }
51
52 [-] public void setCategory(String category) {
53     |     this.category = category;
54     | }
55
56 [-] public double getPrice() {
57     |     return price;
58     | }
59
60 [-] public void setPrice(double price) {
61     |     this.price = price;
62     | }
63
64 [-] public int getQuantity() {
65     |     return quantity;
66     | }
67
68 [-] public void setQuantity(int quantity) {
69     |     this.quantity = quantity;
70     | }
```

Output:

## Add New Product

Name:

Category:

Price:

Quantity:

Add Product

---

## Product List

[Add Another Product](#)

| ID | Name | Category | Price | Qty | Action                                        |
|----|------|----------|-------|-----|-----------------------------------------------|
| 6  | Ubat | Vet      | 131.0 | 3   | <a href="#">Delete</a>   <a href="#">Edit</a> |

## Product List

[Add Another Product](#)

| ID | Name       | Category | Price | Qty | Action                                        |
|----|------------|----------|-------|-----|-----------------------------------------------|
| 6  | Ubat       | Vet      | 131.0 | 3   | <a href="#">Delete</a>   <a href="#">Edit</a> |
| 7  | Ubat batuk | Cecair   | 131.0 | 3   | <a href="#">Delete</a>   <a href="#">Edit</a> |